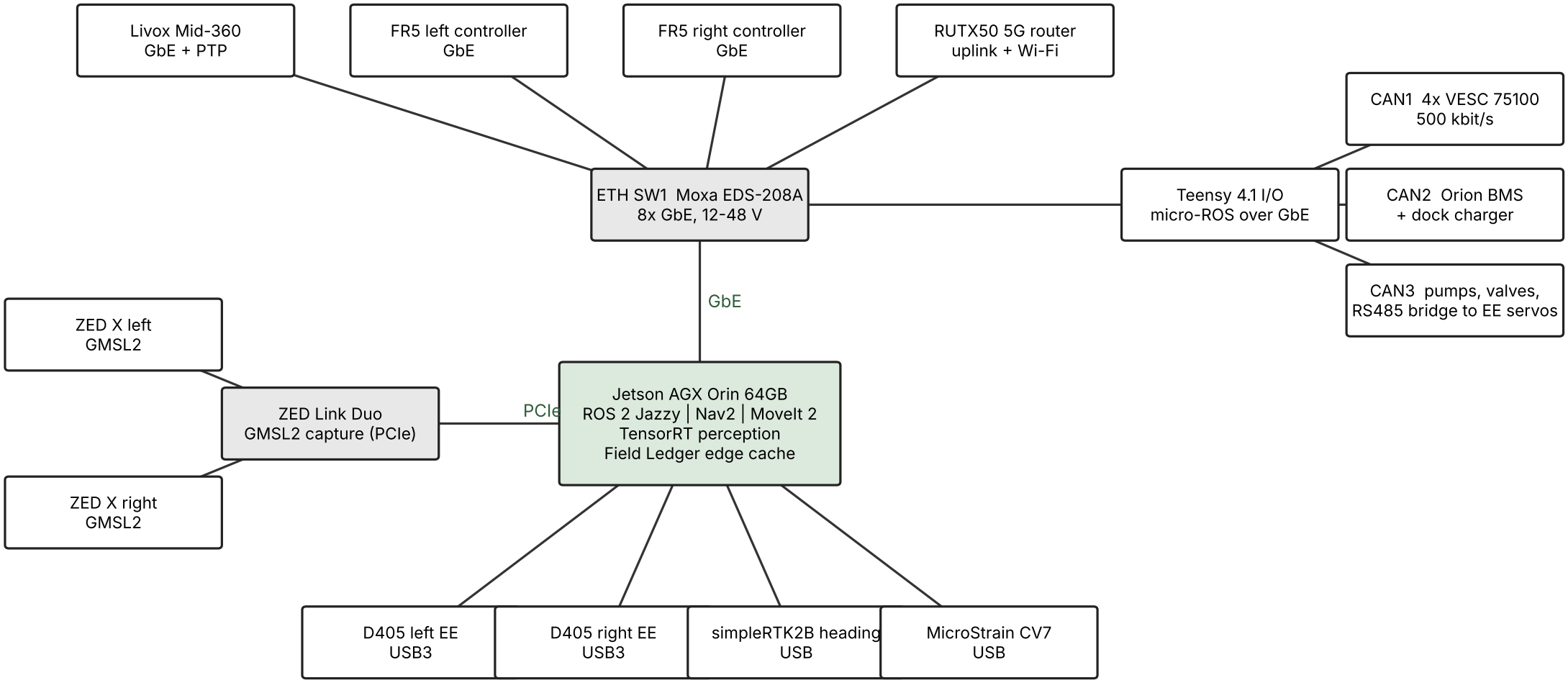
E-03 Data and communications architecture

Ethernet backbone + GMSL2 + USB3 + CAN-FD | ROS 2 Jazzy on Jetson, micro-ROS on Teensy | Agrobo rev A 2026-10-21



RTK corrections: FDOT Florida Permanent Reference Network (FPRN) NTRIP over 5G; fallback = farm base station on a 900 MHz radio.
Time sync: PTP grandmaster on the Jetson to lidar + switch; ZED X pair hardware-triggered by the ZED Link card.
Offline-first: the robot runs its day plan from the local Field Ledger cache; telemetry is buffered and synced when 5G returns.